

Multiple Choice

1. **Answer:** C

Options: A) They are generated when memory cycles are "stolen".; B) They are used in place of data channels.; C) They can indicate completion of an I/O operation.; D) They cannot be generated by arithmetic operations.

Note: Interrupts are signals sent to the processor that notify it when an I/O operation has completed, allowing the CPU to efficiently manage multiple tasks by responding to events as they occur.

2. **Answer:** C

Options: A) It was first proven by Alan Turing.; B) It has not yet been proven, but finding a proof is a subject of active research.; C) It can never be proven.; D) It is now in doubt because of the advent of parallel computers.

Note: Church's thesis is a foundational principle in computability theory that asserts the equivalence of various formal definitions of computable functions, but it cannot be proven because it is not a mathematical statement with a definitive truth value, rather it is a hypothesis about the nature of computation.

3. **Answer:** A

Options: A) A group of cookies stored by the user's Web browser; B) The Internet Protocol (IP) address of the user's computer; C) The user's e-mail address; D) The user's public key used for encryption

Note: A group of cookies stored by the user's Web browser can track and store extensive information about a user's online behavior and preferences, making them a significant threat to personal privacy compared to other options.

4. **Answer:** A

Options: A) 1; B) 2; C) 3; D) 4

Note: In Python 3, the 'min()' function returns the smallest item from an iterable, and since 1 is the smallest value in the list $l = [1, 2, 3, 4]$, it is the correct answer.

5. **Answer:** D

Options: A) 11111111, 00000001; B) 00000001, 10000000; C) 11111111, 10000001; D) 10000001, 10101010

Note: The addition of the two's-complement numbers 10000001 (which represents -127) and 10101010 (which represents -86) results in a sum of 00101011 (which represents 43), causing an overflow because the sign bits of both operands are 1, indicating negative numbers, while the result is positive.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because Python list slicing uses zero-based indexing, so the slice 'list[1:3]' retrieves the elements at indices 1 and 2, which correspond to 786 and 2.23 in the given list.

7. **Answer:** True

Options: A) True; B) False

Note: The expression evaluates to 1 because the modulus operator (%) returns the remainder of 3 divided by 3, which is 0, and adding this to 1 results in 1.

8. **Answer:** False

Options: A) True; B) False

Note: Static allocation is insufficient for managing memory in recursive procedures because the depth and number of active recursive calls can vary unpredictably, requiring dynamic allocation to accommodate fluctuating memory needs.

9. **Answer:** True

Options: A) True; B) False

Note: The language $\{ ww \mid w \text{ in } (0 + 1)^* \}$ is not context-free, as it requires the ability to match an arbitrary number of symbols in two separate segments, which is beyond the capabilities of pushdown automata, while Turing machines can recognize it due to their more powerful computational model.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the built-in max() function in Python 3 is specifically designed to iterate through the elements of a list and return the one with the highest value based on the default comparison criteria.

Long Form Response

11. **Answer:** def area_tetrahedron(side): | return area

Note: The answer correctly outlines the need for a function that calculates the area of a tetrahedron based on its side length, indicating an understanding of both function creation and the geometric properties of the tetrahedron.

12. **Answer:** def split_list(text): | return (re.findall('[A-Z][^A-Z]*', text))

Note: correct because it utilizes the regex pattern '[A-Z][^A-Z]*' to effectively match and extract substrings of uppercase letters, thereby splitting the input string at each uppercase letter.

End of Answer Key